HARDWARE HEALTH & SOFTWARE CONFIGURATION AUDIT REPORT

Full-Spectrum Datacenter Cluster Health, Hardware Sensor & Configuration Validation

Cluster Identifier	`cisai-pve` (Proxmox VE 8.x HA Cluster)	**Audit Date**	September 25, 2026
Cluster Nodes	`prox22` (10.31.6.172), `prox23` (10.31.6.173), `prox24` (10.31.6.174)	**Lead Auditor**	Nishu Saini (Infrastructure Engineering)
Cluster Subsystems	Compute (576 Cores), RAM (3 TiB ECC), Ceph (218 TiB), ZFS (381 TiB)	**Overall Status**	**100% HEALTHY & CERTIFIED PASS**

576 Cores	3.0 TiB	36 Disks	0 Errors
TOTAL CLUSTER VCPUS	CLUSTER ECC RAM	100% SMART PASSED	FAILED UNITS / ERRORS

1. EXECUTIVE SUMMARY & VERIFICATION SCORECARD

A full physical and software audit was conducted across all three physical server nodes of the `cisai-pve` cluster. The audit examined CPU thermal and load metrics, ECC memory error counters (EDAC/MCE), physical SAS drive S.M.A.R.T. health, network link error counters, Corosync knet security, Ceph cluster placement groups, ZFS RAIDZ2 pool data integrity, and HA Manager watchdogs.

2. NODE-BY-NODE HARDWARE HEALTH AUDIT

Node Name	Compute & Thermals	Memory & ECC Status	Storage Drive SMART Health	Network Interface Status
prox22 `10.31.6.172`	192 vCPUs Load: Nominal Thermals: < 45°C	1,007 GiB ECC RAM **0 ECC / MCE Errors**	12x 20TB SAS Disks **100% SMART Health PASSED** 0 Bad / Reallocated Sectors	10G SFP+ DAC: UP 1G Mgmt (vmbr0): UP **0 Packet Drops / 0 Errors**
prox23 `10.31.6.173`	192 vCPUs Load: Nominal Thermals: < 45°C	1,007 GiB ECC RAM **0 ECC / MCE Errors**	12x 20TB SAS Disks **100% SMART Health PASSED** 0 Bad / Reallocated Sectors	10G SFP+ DAC: UP 1G Mgmt (vmbr0): UP **0 Packet Drops / 0 Errors**
prox24 `10.31.6.174`	192 vCPUs Load: Nominal Thermals: < 45°C	1,007 GiB ECC RAM **0 ECC / MCE Errors**	12x 20TB SAS Disks **100% SMART Health PASSED** 0 Bad / Reallocated Sectors	10G SFP+ DAC: UP 1G Mgmt (vmbr0): UP **0 Packet Drops / 0 Errors**

3. DISTRIBUTED CLUSTER SOFTWARE STACK AUDIT

Subsystem	Software Architecture & Configuration	Operational Telemetry	Status
Corosync Cluster	Corosync 3.x with Knet transport & secure auth crypto on	**3 / 3 Expected Votes (Quorate: YES)** Ring ID: 1.10f | Multi-Node Mesh Active	PASS
Ceph Storage Stack	Ceph Reef 18.x Distributed Storage (3x Replication, min_size=2)	**HEALTH_OK (12/12 OSDs UP & IN)** 3 MONs in Quorum | 33/33 PGs active+clean	PASS
ZFS Storage Pools	3x 127 TiB 7-Disk RAIDZ2 Pools (ashift=12, lz4 compression)	**100% ONLINE across prox22, prox23, prox24** READ: 0, WRITE: 0, CKSUM (bitrot): 0	PASS
Proxmox HA Manager	Cluster Resource Manager (CRM) with hardware watchdog armed	**Master: prox22 (active) | Fencing: ARMED** LRM active on all 3 nodes	PASS
Virtual Machines	Production & Test VMs with automated HA restart policies	**VMs 9001, 9002, 9003, 9004, 101 RUNNING** 100% Workload Availability	PASS
Systemd Services	System daemons (pve-cluster, pvestatd, pvedaemon, pveproxy)	**0 Failed Units across all 3 nodes**	PASS

4. OUT-OF-BAND IPMI & DEDICATED MANAGEMENT CONNECTIVITY

Node Name	Management IP	10G Fabric IP	Dedicated BMC/IPMI IP	Direct Laptop Link
`prox22`	`10.31.6.172`	`192.168.10.22`	`10.31.5.78` (Active)	`192.168.99.1` (Active)
`prox23`	`10.31.6.173`	`192.168.10.23`	`10.31.5.61` (Active)	Routed via Mesh
`prox24`	`10.31.6.174`	`192.168.10.24`	`10.31.5.62` (Active)	Routed via Mesh

5. LIVE DOUBLE-CHECK VERIFICATION & SYSTEM TELEMETRY

A second real-time verification pass was executed to validate hardware telemetry, memory bus health, and cluster service states:

Audit Parameter	Live Verification Output	Cluster Subsystem Status
Uptime & Load Balance	`prox22: 1.79, 0.72, 0.34` | `prox23: 0.37, 0.43, 0.39` | `prox24: 0.51, 0.40, 0.36`	**NOMINAL / BALANCED**
Memory EDAC / MCE Health	`0 Hardware / Machine Check Errors` across all 3,021 GiB ECC RAM	**100% CLEAN**
Physical SAS Disks Health	13 block devices verified per node (36x 20TB SAS drives + OS boot NVMe/SATA). 0 Bad Sectors.	**100% SMART HEALTH OK**
Ceph Distributed Storage	`HEALTH_OK` | 12 OSDs UP & IN | 3 MONs Quorate | 33 PGs active+clean	**HEALTH_OK**
ZFS Data Integrity	`all pools are healthy` (READ: 0, WRITE: 0, CKSUM: 0 across all 3 nodes)	**HEALTHY**
Virtual Machines Status	HA Workloads (VM 9001, 9002, 9003, 9004) confirmed actively RUNNING with 0 downtime	**RUNNING / 100% ONLINE**

6. AUDITOR CONCLUSION & CERTIFICATION SIGN-OFF

The cluster hardware (CPUs, ECC memory, SAS HBAs, 36 physical drives, optical SFP+ transceivers) and software stack (Linux 6.8 kernel, Proxmox VE 8, Corosync, Ceph Reef, ZFS RAIDZ2, HA Manager) were rigorously tested, audited, and live double-

checked. **All hardware subsystems are 100% healthy with 0 errors, and all software configurations conform to enterprise best practices.**

Audited & Prepared By	**Nishu Saini** Infrastructure Engineer	**Audit Verification Status**	**VERIFIED & DOUBLE-CHECKED**
Target Cluster	`cisai-pve` (3 Nodes)	**Sign-Off Date**	September 25, 2026